

VASUDEV BANSAL

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EDUCATION

2024 – 2028	B.Tech in Computer Science Engineering, USICT GGSIPU Delhi	CGPA: 8.1
2023	Class XII (CBSE), S.D. Public School	92%
2021	Class X (CBSE), S.D. Public School	91%

PROFILE

Computer Science undergraduate with experience in Full-Stack and Backend Development, building scalable web applications, AI-powered platforms and REST APIs. Currently working as a Software Development Engineer Intern at ChalksnBoard, contributing to both the company's production website and AI-powered communication systems using React, Next.js, Node.js, Docker and Redis.

EXPERIENCE

Software Development Engineer (Full Stack) Intern — ChalksnBoard June 2026 – Present  chalksnboard.in

- Spearheaded the modernization of the **ChalksnBoard** website by migrating it to a scalable **React.js** and **Next.js** architecture, improving maintainability and brand presence.
- Enhanced user experience by implementing interactive scroll animations, responsive layouts and immersive web experiences using **Framer Motion**, **GSAP**, **Three.js** and **Tailwind CSS**.
- Designed reusable React components and modular UI sections, accelerating the development of landing pages and new website features.
- Improved website performance and technical SEO by leveraging **Next.js** server-side rendering, asset optimization and responsive design best practices.
- Developing **Vasooli AI**, an AI-powered fee collection platform using **Node.js**, **Express.js**, **MongoDB** and secure **JWT Authentication**.
- Built scalable **REST APIs** and integrated **WhatsApp (Interakt)**, **Twilio SMS**, **Redis**, **BullMQ** and Cron Jobs for automated communication workflows.
- Containerized backend services using **Docker** and explored multilingual AI voice technologies including **Microsoft Edge TTS**, **Coqui XTTS**, **ElevenLabs** and **LiveKit**.

PROJECTS

Vasooli AI – AI Powered Fee Collection Platform

- Built an AI-powered fee collection platform using Node.js, Express.js, MongoDB and React Native.
- Developed secure JWT Authentication, REST APIs and role-based authorization.
- Integrated WhatsApp APIs, Twilio SMS, Redis and BullMQ.
- Used Docker for deployment and explored Microsoft Edge TTS, Coqui XTTS, ElevenLabs and LiveKit.

Spotify Backend Clone

- Developed a scalable music streaming backend using Node.js, Express.js, MongoDB Atlas and JWT Authentication.
- Implemented authentication, playlist management, song uploads and streaming REST APIs.
- Designed optimized MongoDB schemas and secure middleware architecture.
- Integrated Cloudinary for media storage.

AI Resume Match Score Analyzer

- Developed an AI-powered resume analyzer using Gemini API.
- Implemented ATS score calculation, skill-gap analysis and roadmap generation.
- Built backend APIs and responsive frontend interfaces.

AssetFlow – Enterprise Asset & Resource Management System

 [Live Demo](#)

- Designed and deployed a full-stack enterprise asset management platform using **Next.js**, **PostgreSQL**, **Prisma ORM** and **Supabase** to centralize asset tracking, maintenance and resource allocation.
- Implemented secure **JWT Authentication** with strict **Role-Based Access Control (RBAC)** supporting four user roles: Admin, Manager, Department Head and Employee.
- Engineered a conflict-free shared resource booking system with approval workflows, transfer requests and calendar scheduling to eliminate double-booking.
- Built scalable **REST APIs**, real-time KPI dashboards and optimized database operations using **Next.js Route Handlers** and **Prisma ORM**.
- Developed a responsive enterprise-grade UI using **CSS Modules**, Glassmorphism design principles and an Odoo-inspired design system.

TECHNICAL SKILLS

Languages	C++, Java, Python, JavaScript, SQL, HTML/CSS
Frontend Development	React.js, Next.js, TypeScript, Tailwind CSS, HTML5, CSS3, JavaScript, React Native (Expo), Framer Motion, GSAP, Three.js
Backend Development	Node.js, Express.js, REST APIs, JWT Authentication, Middleware, API Integration, Prisma ORM, BullMQ, Cron Jobs
Databases	PostgreSQL, MongoDB Atlas, MySQL, SQLite, Redis, Supabase
AI Technologies	Gemini API, Microsoft Edge TTS, Coqui XTTS, ElevenLabs, LiveKit
DevOps & Tools	Docker, Git, GitHub, Postman, Render, Vercel, Redis, BullMQ, Cron Jobs

ACHIEVEMENTS

- Solved **250+** LeetCode problems across Arrays, Trees, Graphs, Dynamic Programming and Binary Search.
- Built multiple production-ready full-stack and backend applications using React, Next.js, Node.js, MongoDB, PostgreSQL and RESTful APIs.
- Hands-on experience with Docker, Redis, BullMQ, AI voice technologies and scalable backend architecture.
- Strong interest in Backend Engineering, Distributed Systems and AI-powered applications.